

AI 流程自动化采纳机制：最终核验结果已经闭环。

基于中小企业 AI 流程自动化采纳机制研究

从“待核验训练路线”升级为“数据、模型、Agent 已验证”的课堂答辩版本。

544

Stage 1 可建模样本

5,814

Stage 2 可建模样本

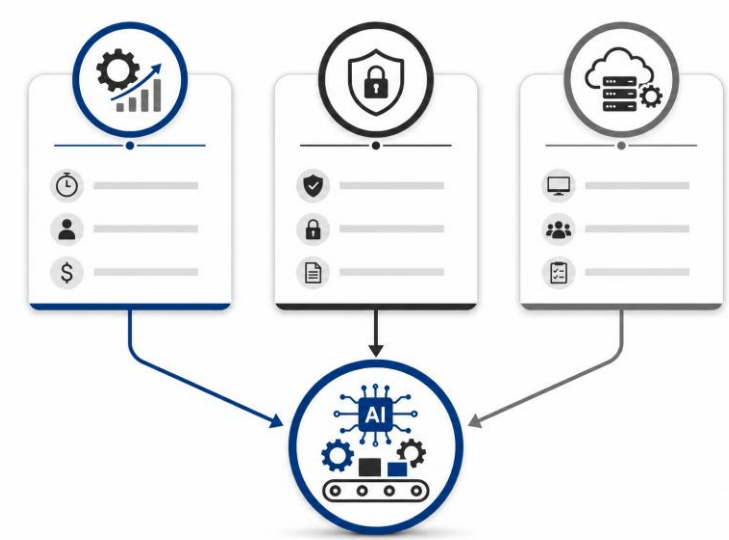
0.7137

Stage 2 GroupKFold R²

0.0

Agent hallucination rate

口径锁定：Stage 1 = SME 机制解释；Stage 2 = GE10 行业/区域外部验证。



两阶段设计把中小企业机制解释和外部泛化验证分开。

Stage 1

SME mechanism interpretation layer

553

544

36

面板行数

可建模样本

geo 分组

年份范围：2021 - 2025

研究用途：中小企业规模层机制分析

``data/processed/eurostat_ai_panel.csv``

Stage 2

GE10 industry and region external validation layer

5,814

5,814

36

面板行数

可建模样本

geo 分组

年份范围：2021 - 2025

行业分组：50 个 NACE

``data/processed/stage2_industry_panel.csv``

What the chart shows

Stage 1 answers the original SME mechanism question; Stage 2 checks whether the pattern remains stable in broader official statistics.

Why it matters

This prevents overstating Stage 2 as SME-only while still using its scale for external validation.

Decision it supports

最终模型页只展示已核验指标，不再混用历史调参分数。

Layer	Champion	GroupKFold R ²	Time holdout R ²	Industry holdout R ²
Stage 1	ridge	0.8680	0.9312	N/A
Stage 2	extra_trees	0.7137	0.7019	0.9521

0.8680

Stage 1 Ridge GroupKFold R²

0.7137

Stage 2 ExtraTrees GroupKFold R²

0.9521

Stage 2 industry holdout R²

What the chart shows

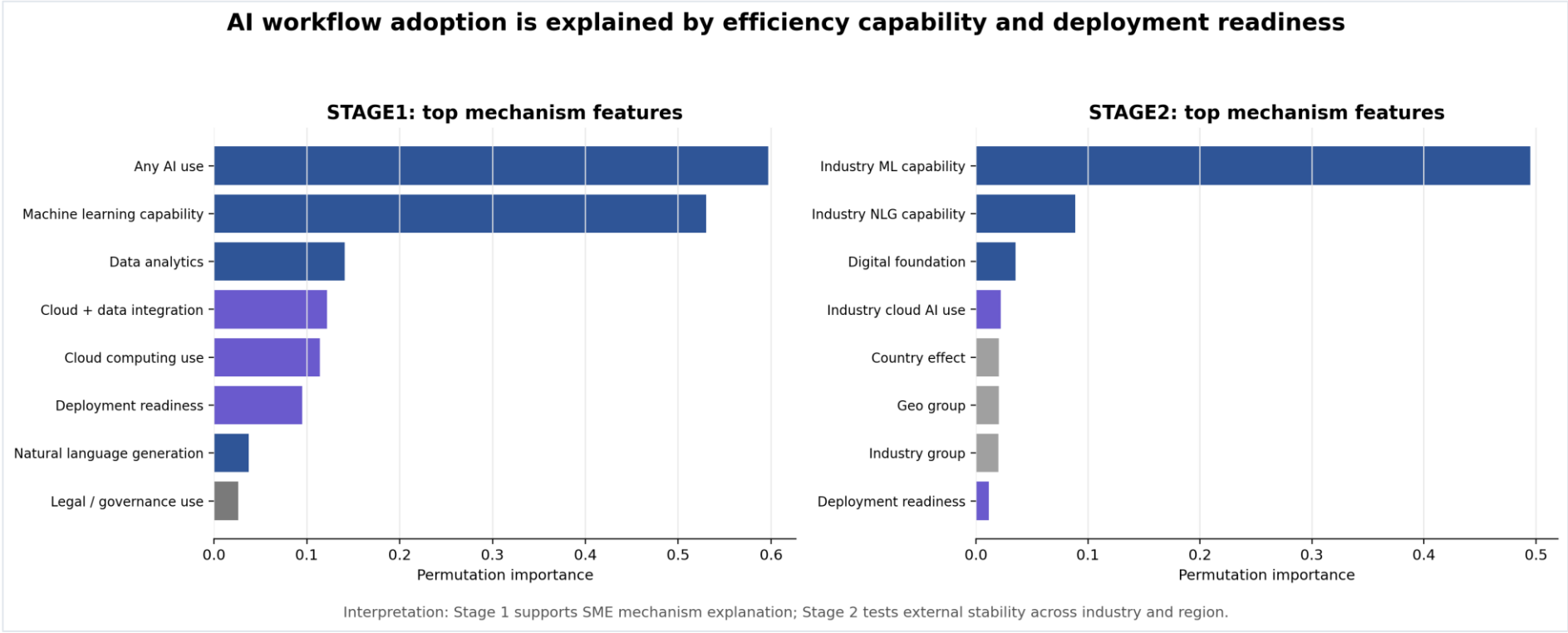
Ridge is retained for Stage 1 mechanism explanation; ExtraTrees is the Stage 2 champion.

Why it matters

GroupKFold by geo is more convincing than random splits because it tests cross-region transfer.

Decision it supports

双阶段特征重要性把模型结果重新解释回研究机制。



What the chart shows

Stage 1 is driven by AI use, ML capability and deployment readiness.

Why it matters

Stage 2 repeats the ML capability signal while adding industry and region controls.

Decision it supports

This supports the claim that adoption depends on capability and readiness, not generic AI enthusiasm.

研究 Agent 的价值是证据优先，而不是凭空生成答案。

6

evaluation cases

1.0

tool success rate

1.0

citation accuracy proxy

0.0

hallucination rate

0.0436s

average latency

Guardrail

- 数值型回答必须来自工具、模型 API 或 evidence files。
- 超范围问题返回“无法确认”，不向中国所有中小企业做无证据外推。

Tool stack

- query_indicator / predict_adoption / cite_source / recommend_deployment
- RAG 只索引经过验证的报告、模型结果和来源说明。

What the chart shows

The Agent is tested with metric lookup, prediction, citation, deployment and unknown-claim cases.

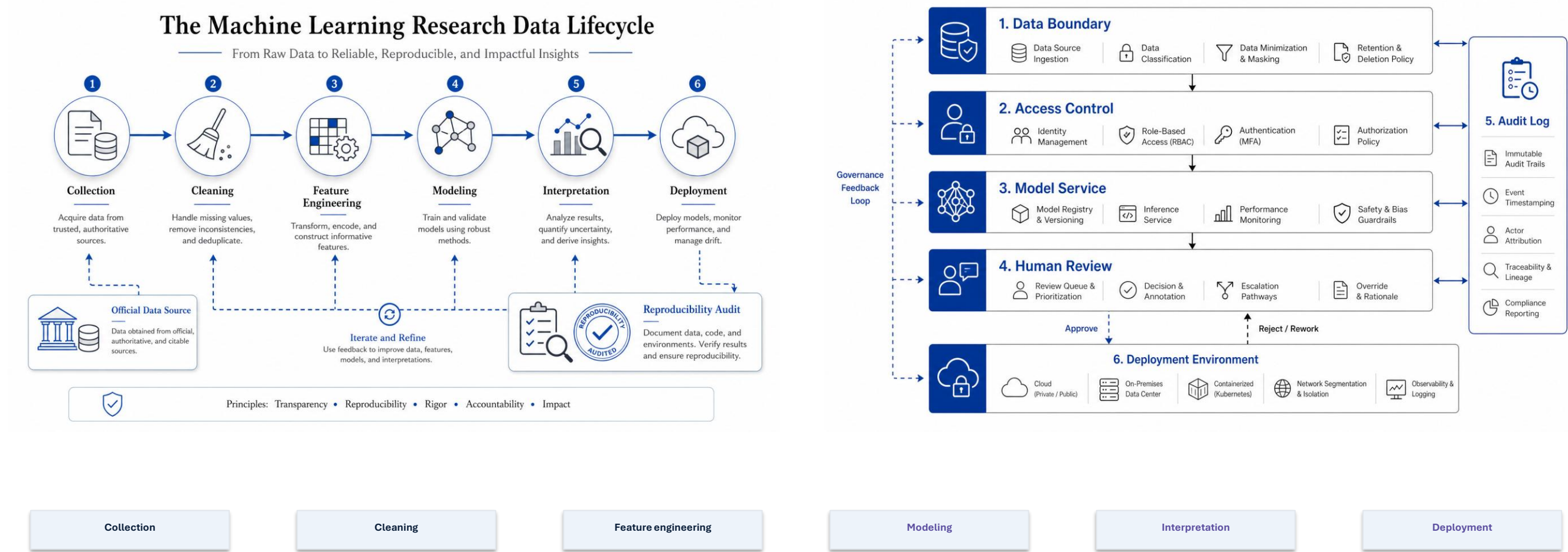
Why it matters

The zero hallucination proxy comes from the explicit refusal behavior on unsupported claims.

Decision it supports

This makes aizhijia tech a research-backed operating layer, not only a demo website.

最终版本把数据生命周期、安全边界和复现证据放在同一条链路里。



What the chart shows

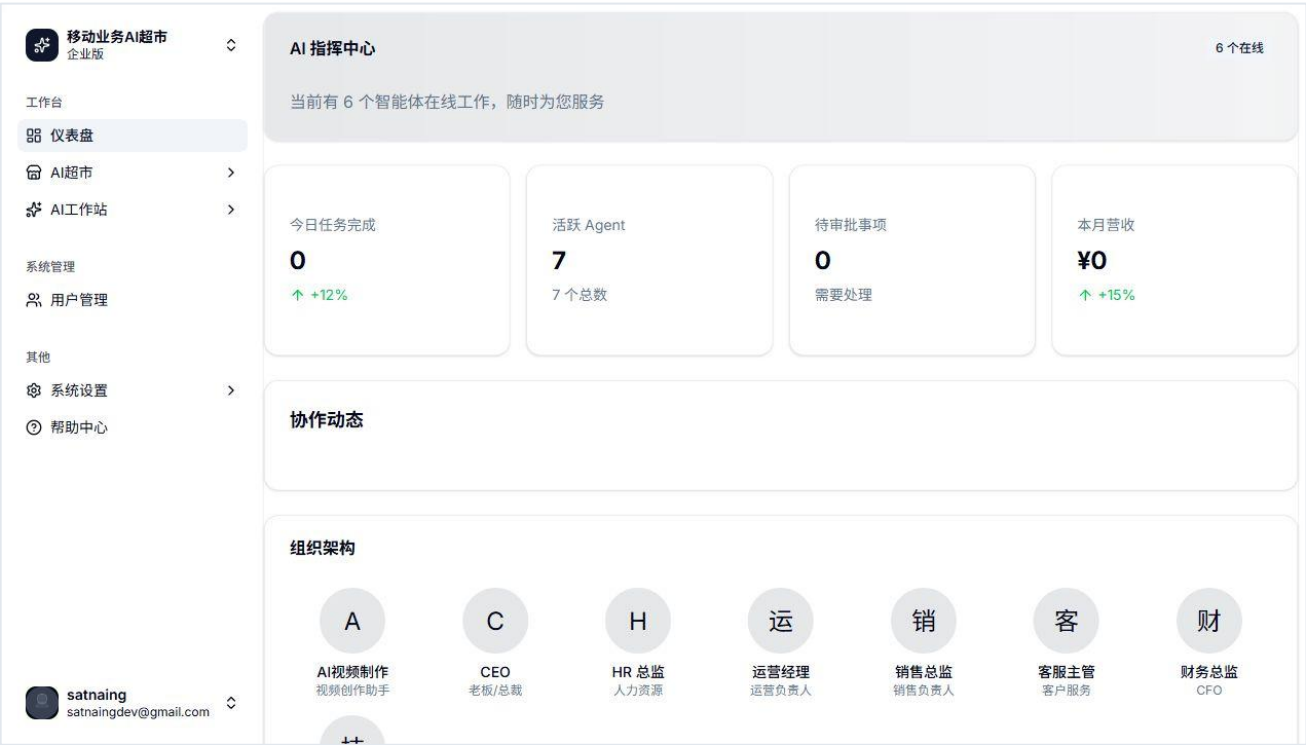
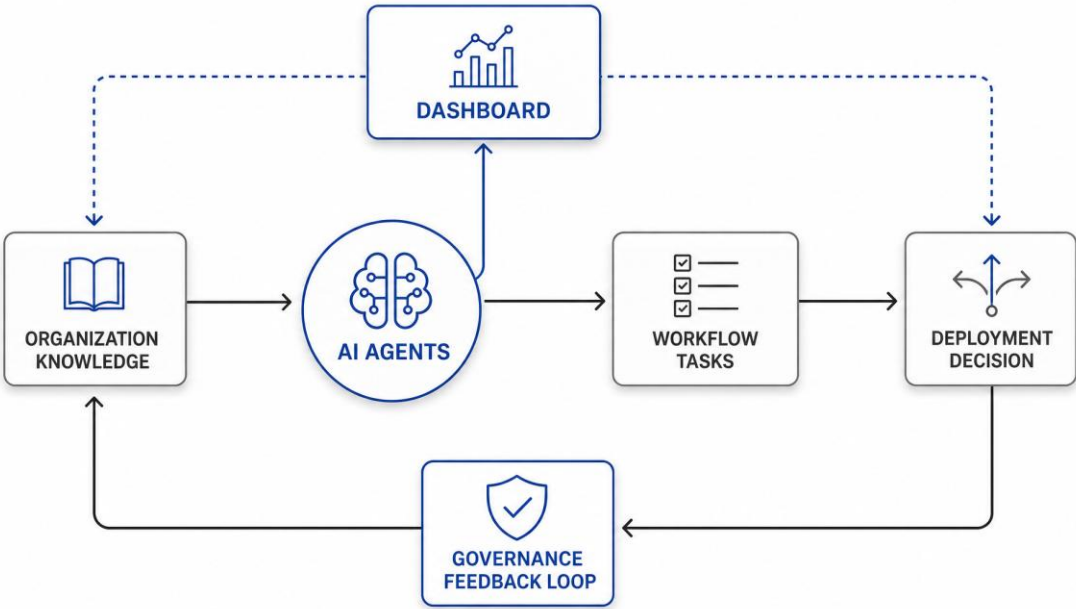
The digital lifecycle is visible from official acquisition to deployment recommendation.

Why it matters

Security is not a side issue; it changes which deployment mode is appropriate.

Decision it supports

ai.zhjjq.tech 把模型结论落到真实 AI 办公流程中。



Research model → enterprise profile → deployment recommendation → AI workstation workflow

What the chart shows

The website is not just a screen; it is the operating surface for workflow agents and governance feedback.

Why it matters

The model helps decide whether an enterprise should use SaaS, API, local or hybrid deployment.

Decision it supports

Source: final research validation pack, e789173

This links the machine-learning case to a practical SME AI automation product scenario.

这份结题案例已经形成“数据真实、模型严谨、Agent 可落地”的闭环。

01	Official data	Eurostat source files, manifest hash audit and reproducible panels.
02	Machine learning	Ridge, ExtraTrees, GroupKFold, time holdout and industry holdout.
03	Explainable mechanism	Efficiency demand × security concern × deployment readiness.
04	Agent deployment	Tool-based answers, evidence files and unable-to-confirm guardrail.

下一步只需要把国内问卷和真实 workflow 日志接入，就能继续增强中国中小企业场景解释力。